

Davidson Blessons

A NotebookLM-powered Bible study app | Project report, October 2026

In one sentence: Davidson Blessons lets you browse all 66 books of the Bible chapter by chapter, and for each chapter shows the NIV text plus plain-language study notes (meaning, three key takeaways, and a reflection question), retrieved live from the owner's own Google NotebookLM notebook.

Links

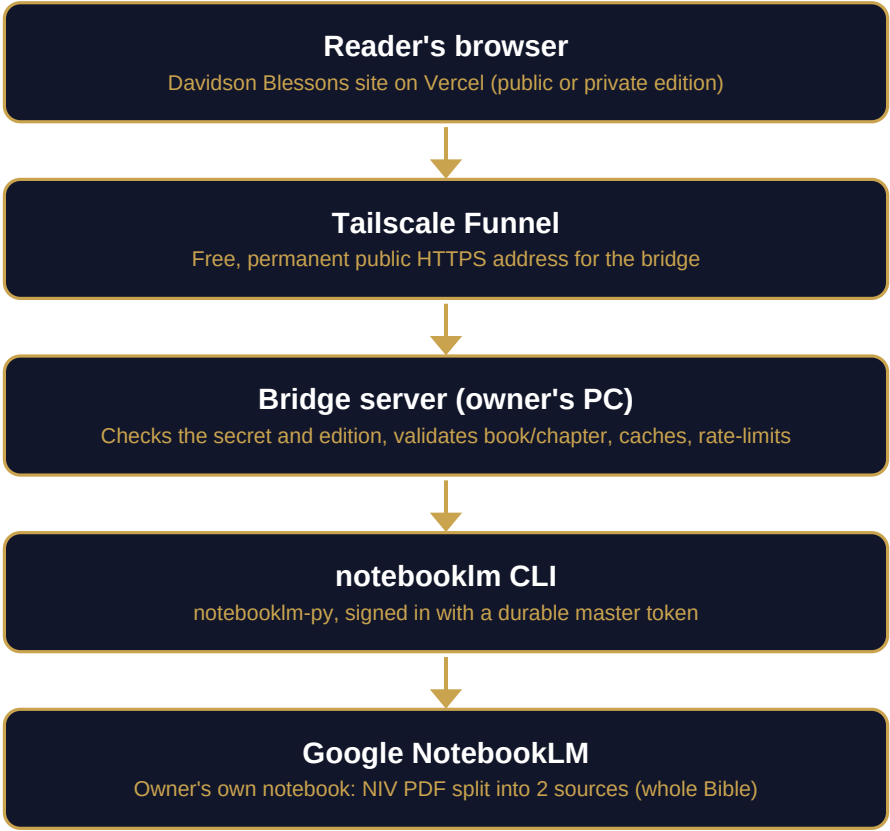
What	Where
Public site (anyone with the link)	https://davison-blesson.vercel.app
Source code	https://github.com/JamieTurner2026/Davidson-Blessons

At a glance

Area	Status
App: 66-book, chapter-by-chapter navigation	Done, live on Vercel
Live chapter text + study notes from NotebookLM	Done, works end-to-end on the owner's PC
Whole-Bible coverage in the notebook	Done: all 1,189 chapters
Public (excerpt) and private (full) editions	Done, both deployed
Security review with live tests	Done, 2 issues found and fixed
Auto-start and self-healing bridge server	Done, confirmed after a real restart
Code on GitHub	Done
Free tunnel so the live sites reach the bridge	Done: Tailscale Funnel, no cost

How it works

When a reader opens a chapter, the site asks a small **bridge server** running on the owner's own computer. The bridge asks the owner's **NotebookLM notebook** two questions (the chapter's text, and an explanation of it) and sends both answers back. The owner's Google sign-in never leaves their computer. A free **Tailscale Funnel** gives the bridge a permanent public web address so the Vercel-hosted sites can reach it.



The two editions

	Public edition	Private edition
Who can open it	Anyone with the link	Only the owner (Vercel sign-in required)
Scripture shown	A short excerpt: at most 15 words of verse 1, with attribution	Full chapter, verse by verse
Study notes	Full AI study notes, written in plain language	Full AI study notes
Limits	10 new chapter loads per visitor per hour	None

Why two editions? The NIV is copyrighted. The full text comes from the owner's own copy for personal study, so full chapters are only shown in the owner's private edition. The public edition shows a short excerpt, and that limit is enforced by the bridge server itself, not just the web page, so it holds even if someone calls the bridge directly.

What was built

1. The web app (React + Vite + Tailwind)

- Rebuilt from an earlier 37-lesson version into full navigation: Old/New Testament, then book, then chapter, for all 66 books and 1,189 chapters.
- Chapter view with two clearly separated sections: **NIV Scripture** and **Learning Notes**.
- Loaded chapters are saved in the browser, so revisiting one is instant. "Mark this chapter learned" progress is saved too (the original app lost everything on refresh).
- Honest states: a loading indicator, a clear message when a chapter isn't available, and never any made-up text if something fails.
- Candlelit design kept from the original: deep navy, gold accents, parchment Scripture card.

2. The bridge server (Python, on the owner's PC)

- Small web server that takes a book and chapter, asks NotebookLM, and returns the answer.
- Signs in to NotebookLM with a durable **master token**, so it keeps working without anyone signing in again.
- Separate secrets for the public and private editions; everything else is turned away.
- Saves each finished chapter to disk, so each chapter only has to be fetched from NotebookLM once.
- Accepts connections only from the computer it runs on.
- Starts automatically at sign-in, and restarts itself within about 10 seconds if it crashes.

3. The NotebookLM notebook

- The owner's own notebook, holding the owner's own NIV PDF.
- Coverage originally stopped at Isaiah 25 because NotebookLM only reads about 500,000 words per source and the full Bible is about 729,000 words. The PDF was split in two (Genesis to Isaiah 25, and Isaiah 26 to Revelation), which fixed it: **all 1,189 chapters are covered**.

4. Hosting and source code

- Deployed on Vercel as two editions from one project.
- Code published to GitHub, with a check beforehand that no secrets were included.

Security review

Before anything went public, the bridge server was tested with real attack-style requests. Two genuine problems were found and fixed, and every fix was re-tested.

Finding	Risk	Fix (verified)
If the server's secret setting was missing, it served chapters to anyone with no password at all ("fail-open").	High	Now refuses every request if a secret is missing, and compares secrets in a way that doesn't leak timing hints.
Bad input such as chapter "abc" went straight to NotebookLM (about a 52-second wasted query) and let arbitrary text reach the AI prompt.	Medium	Every book and chapter is checked against the real 66-book list first. Bad requests are rejected in about 0.2 seconds.
The server was reachable by any device on the home network.	Low	Now only accepts connections from the owner's own computer.
Once public, strangers could use up the owner's NotebookLM quota.	Low	Public edition limited to 10 new chapter loads per visitor per hour.

Also confirmed safe: requests from websites other than the app are blocked, and the public site's code does not contain the private edition's secret.

Accepted by design

The public edition's secret is necessarily visible in the public site's code. That's acceptable because the bridge only ever gives that secret a short excerpt, and it is rate-limited.

Problems solved along the way

Problem	Cause	Solution
NotebookLM stopped answering after a few days	The saved browser sign-in expired	Switched to master-token sign-in, which renews itself
Server crashed on some answers	Windows read the text in the wrong character set	Read all output as UTF-8
Chapters opened close together seemed to hang	One slow request blocked all the others	Requests now wait in an orderly queue while the server stays responsive
Auto-start with Windows Task Scheduler never worked	Its sign-in trigger didn't fire reliably on this machine	Replaced with a Startup-folder launcher that restarts the server if it crashes
Only Genesis to Isaiah 25 was available	NotebookLM's limit of about 500,000 words per source	Split the PDF into two sources
A tunnel normally needs a paid domain	Cloudflare's named tunnels require owning a domain	Used Tailscale Funnel: free, with a permanent address
Chat-style "Would you like me to...?" lines in answers	NotebookLM adds follow-up offers	Stripped automatically before display

Directions considered and set aside

- **YouVersion's official NIV API:** requires partner approval, so it wasn't available.

- **Hand-writing lessons for all 1,189 chapters:** replaced by live retrieval from NotebookLM.
- **A shared, third-party full-NIV notebook:** deliberately not used, because its creator's right to distribute the full text couldn't be confirmed. The project uses only the owner's own copy.

Status and next steps

Everything is connected

All planned steps are done. Both editions on Vercel now load chapters live through the bridge, the bridge and tunnel both start automatically when the PC starts, and the notebook covers all 1,189 chapters.

Possible future improvements

- Move the bridge to an always-on server so chapters load even when the owner's PC is off.
- Connect the GitHub repository to Vercel so every code change deploys automatically.

Good to know

- Because the bridge runs on the owner's PC, chapters only load while that PC is on, awake, and signed in.
- The NIV's permission terms allow quoting up to 500 verses without written permission. The public edition shows a slice of one verse per chapter, but across all 1,189 chapters that could add up past 500, so keep this in mind before promoting the public site widely.

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